

Chapter 4

Present Worth and Related Economic Criteria

So far, we have established methods that allow transforming various cash flows by using interest rates. In the forthcoming chapters, we develop and demonstrate basic techniques to **evaluate feasible alternatives**. Ultimately, and as we have seen in Section ??, Engineering Economy attempts to identify the *best* (most viable) alternative(s) from an *economic perspective*. An economic perspective requires quantitative criteria for decision-making.

This chapter studies the Present and Future Worth criteria. In particular, we develop techniques for comparing two or more alternatives using the Present Worth and Future Worth criterion.

Before delving into these methods, we discuss characteristics of alternatives.

4.1 Alternatives

As shown in Figure 4.1, proposals start from ideas, experience, brainstorming, etc., and are subject to an initial filtering process that disregards non-feasible ones. A viable proposal is called an alternative. Afterwards, we recognize two types of economic projects: Mutually Exclusive and Independent projects. The Do-Nothing (DN) alternative is also a feasible alternative. Finally, before performing the Present Worth analysis, we identify two types of cash flows: Revenue and Cost (or Service).

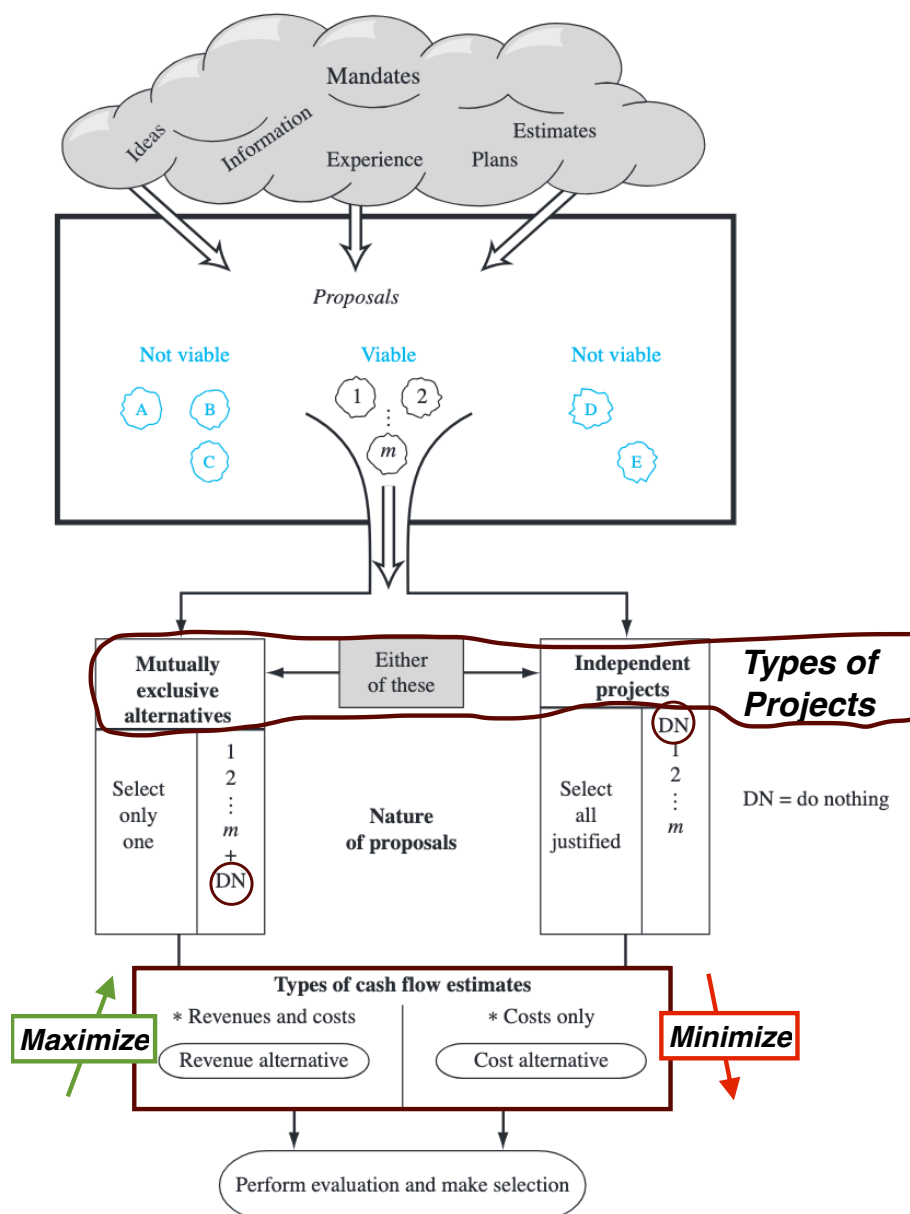


Figure 4.1: Progression from proposals (alternatives) to economic evaluation to selection. The two different types of projects and cash flows are highlighted *source: blank engineering 2012*

4.1.1 Types of Economic Projects

There are two types of alternatives: Mutually Exclusive alternatives, and Independent projects.

Mutually Exclusive Alternatives:

- Only one of the alternatives can be selected.
- Alternatives compete with one another.

Independent Projects:

- More than one alternative can be selected.
- In the absence of a budget constraint, choose all alternatives that do better than the Do-Nothing alternative.
- Independent projects are competing with the Do-Nothing project.

4.1.2 Cash flow types

- Revenue: Each alternative generates cost (cash outflow) and revenue (cash inflow) estimates, and possibly savings (also considered cash inflows). Revenues can vary for each alternative.

Criteria: Select the alternative that **maximizes** the economic measure of merit, which is profit-based.

- Cost: Each alternative has only cost cash flow estimates. Revenues or savings are assumed equal for all alternatives; thus they are not dependent upon the alternative selected. These are also referred to as service alternatives.

Criteria: Select the alternative that **minimizes** the economic measure of merit, which is cost-based.

Example 4.1. A baby diaper production facility has one production and packaging line producing 200,000 diapers per day with 95% efficiency. The manager wants to improve efficiency; for this, there are many possibilities. Some of these possibilities are listed below.

They can salvage the current production line and buy a new one with higher output and efficiency. The possibilities are:

- A- Buy a new production line with 240,000 throughput and 98% efficiency for 6 million USD. The increased performance will generate an extra revenue of 5,000 USD per day.
- B- Buy a used production line with 200,000 throughput and 96% efficiency for 2.5

million USD. The increased performance will generate an extra revenue of 2,000 USD per day.

- C- Buy a new production line from a sister company with 240,000 throughput and 97% efficiency for 5 million USD. The increased performance will generate an extra revenue of 3,000 USD per day. By working with a sister company, you will be supporting in developing the line and building sustainable knowledge. This alternative carries an extra benefit of future knowledge development and technological transfer¹.

Alternatively, they can keep the line and perform some upgrades to improve performance. The possibilities are:

- 1- Install an online inspection system. The cost is 100,000 USD, and 2,000 per year for a software subscription. This system reduces annual scrap material costs by 15,000 USD. This project also improves customer satisfaction because defects cannot reach the finished goods¹.
- 2- Replace existing guiding systems with newer more accurate ones. The cost is around 16,000 USD. This replacement reduces annual maintenance and downtime costs by 5,000 USD.
- 3- Buy a new Cutting die made of tungsten carbide material. The cost of this die is around 300,000 USD. This durable die reduces annual tooling replacement costs by 40,000 USD.

It is clear that Alternatives A, B, and C are mutually exclusive. That is, only one of them can be executed. Hence, we say that these alternatives are competing with each other, as well as with the “do-nothing” alternative (keeping the current line).

On the other hand, alternatives 1, 2, and 3 are independent projects such that the manager can select any combination of them. Another alternative in this case is to do nothing and keep the current line. The do-nothing alternative cannot be combined with the other projects.

¹These are examples of an intangible or irreducible benefit. While such strategic advantages are highly valuable for the company’s long-term growth and competitiveness, they are difficult to quantify economically and translate into a reliable monetary revenue estimate for a Present Worth analysis.

4.2 Present Worth Analysis

Definition – Present Worth (PW) Analysis

Present Worth (PW) Analysis is based on the concept of equivalent worth of all cash flows relative to a base or beginning point in time called the present. All cash inflows and outflows are **discounted** to the present point in time at an interest rate that is generally the Minimum Attractive Rate of Return (MARR).

A positive PW for an investment project represents a dollar amount of profit over the minimum amount required by investors. Furthermore, it is assumed that cash generated by the alternative is available for other uses that earn interest at a rate equal to the MARR. Generally, the higher the Present Worth, the better the alternative.

The PW of future cash flows is found as a function of i as follows:

$$PW(i) = \sum_{k=0}^N F_k(1+i)^{-k}$$

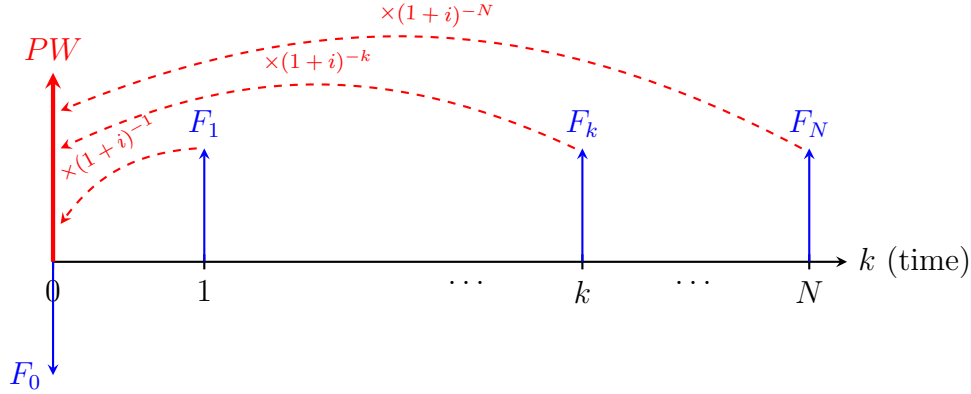


Figure 4.2: Illustrating the process of discounting future cash flows to time 0 to find the Present Worth.

4.2.1 Present Worth Analysis of Equal-Life Alternatives

The PW comparison of alternatives with equal lives is straightforward, but the rule depends on the type of alternatives being evaluated:

- **Mutually Exclusive Alternatives:**

- *Revenue Alternatives:* Select the alternative with the **highest positive** $PW(i = MARR)$. If no alternative has a positive PW, then the Do-Nothing alternative is adopted.

- *Cost Alternatives*: Select the alternative with the least negative (numerically highest) $PW(i = MARR)$. The Do-Nothing alternative is typically not an option here because the service must be provided.
- **Independent Projects**: Select all projects with $PW(i = MARR) \geq 0$, assuming there is no budget constraint. However, in reality there will be constraints on available budget.

Example 4.2. Continuing from the previous diaper production facility example, let us evaluate the alternatives and projects using Present Worth analysis. Assume the organization's MARR is 10% per year². The evaluation period is 5 years, and the facility operates 365 days a year. For simplicity, assume there is no salvage value for any of the alternatives or projects at the end of the 5 years.

The $(P/A, 10\%, 5)$ factor is 3.7908.

Evaluating the Mutually Exclusive (Revenue) Alternatives: These are Revenue Alternatives, so we compute the PW and select the alternative with the highest positive value. The annual revenues are computed by multiplying the daily extra revenue by 365.

- **Alternative A:**

$$PW_A = -6,000,000 + (5,000 \times 365)(P/A, 10\%, 5) = -6,000,000 + 1,825,000(3.7908) = \$918,185.85$$

- **Alternative B:**

$$PW_B = -2,500,000 + (2,000 \times 365)(P/A, 10\%, 5) = -2,500,000 + 730,000(3.7908) = \$267,274.34$$

- **Alternative C:**

$$PW_C = -5,000,000 + (3,000 \times 365)(P/A, 10\%, 5) = -5,000,000 + 1,095,000(3.7908) = -\$849,088.49$$

Decision: Both Alternative A and B yield a positive PW. Alternative A is selected because it yields the highest positive PW ($918,185.85 > 267,274.34$). The Do-Nothing alternative is rejected.

Evaluating the Independent Projects: These independent projects behave as cost-saving alternatives. We select all projects that yield a $PW \geq 0$.

²To connect this to our discussion on bonds in Section ??: if the firm finances this project entirely by issuing bonds, its MARR floor is driven by the current market **Yield to Maturity** of those bonds—not just the coupon rate

- **Project 1:** Cost = \$100,000, Net Annual Savings = \$15,000 – \$2,000 = \$13,000
 $PW_1 = -100,000 + 13,000(3.7908) = -\$50,719.77 \rightarrow \text{Reject } (PW < 0)$
- **Project 2:** Cost = \$16,000, Net Annual Savings = \$5,000
 $PW_2 = -16,000 + 5,000(3.7908) = \$2,953.93 \rightarrow \text{Accept } (PW \geq 0)$
- **Project 3:** Cost = \$300,000, Net Annual Savings = \$40,000
 $PW_3 = -300,000 + 40,000(3.7908) = -\$148,368.53 \rightarrow \text{Reject } (PW < 0)$

Decision: We should only proceed with Project 2, as it is the only independent project that economically justifies its cost at a 10% return rate.

4.2.2 Present Worth Analysis of Different-Life Alternatives

A *fair comparison* of mutually exclusive alternatives requires that PW values represent cash flows associated with equal service. For cost alternatives, failure to compare equal-service will always favor the shorter-lived mutually exclusive alternative, even if it is not the more economical choice, because fewer periods of costs are involved.

Definition – Equal-Service Requirement

The PW of the alternatives **must be compared over the same number of years** and must end at the same time to satisfy the **equal-service requirement**^a.

^aThis is a key assumption when comparing alternatives

The equal-service requirement is satisfied by using either of two approaches:

- **LCM:** evaluate alternatives over the Lowest Common Multiple (LCM) of lives. This method assumes reinvestment at the same cash flow estimates in each life cycle of the LCM planning horizon.
- **Study Period:** Assume a fixed planning horizon (n) and evaluate the alternatives over it. This approach does not necessarily consider the useful life of an alternative.

Note – Independent Projects Evaluation

For independent projects, each project is compared to the do-nothing alternative, and not to each other. Hence, satisfying the equal-service requirement is not a problem. Simply use the MARR to determine the PW over the respective life of each project, and select all projects with a $PW \geq 0$.

Lowest Common Multiple Approach

The LCM method extends the cash flow estimates to the same period. For example, lives of 3 and 4 years are compared over a 12-year period. The initial cost of an alternative is reinvested at the beginning of each life cycle, and the estimated salvage value is accounted for at the end of each life cycle when calculating the PW over the LCM period. This is illustrated in Figure 4.3. The LCM approach requires that the following assumptions be

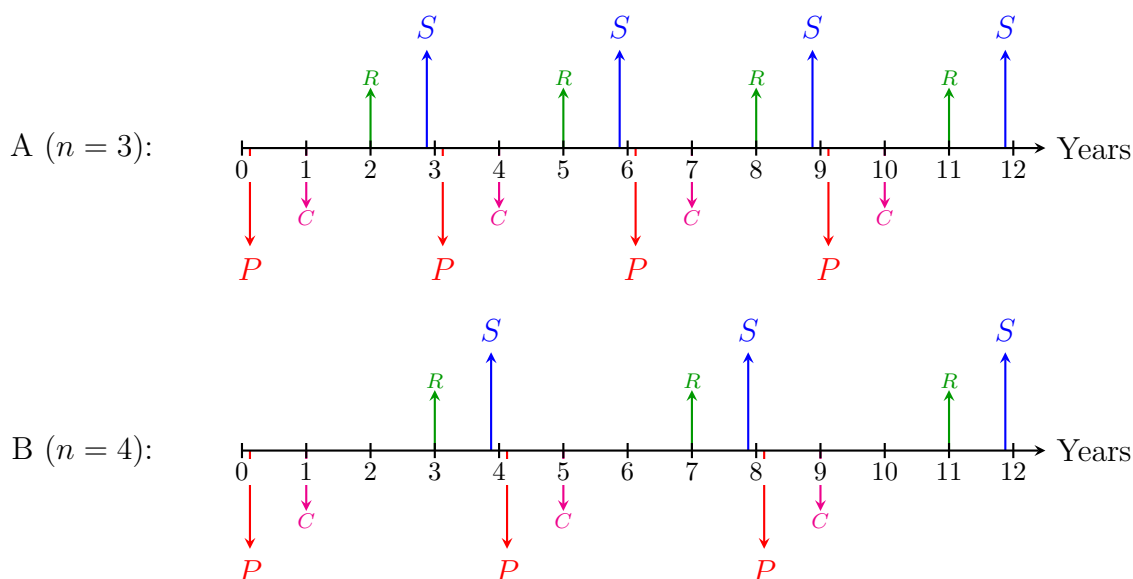


Figure 4.3: Illustration of the LCM approach for alternatives with 3-year and 4-year lives evaluated over a 12-year period. The initial cost (P), the salvage value (S), an intermediate revenue (R) and cost (C) in each cycle are shown.

made about subsequent life cycles:

1. The service provided is needed for LCM years or more.
2. The selected alternative is repeated over each life cycle of the LCM in exactly the same manner.
3. Cash flow estimates are the same in every life cycle.

In fast-moving technology sectors (e.g., semiconductor equipment, EV manufacturing lines), the LCM assumption is frequently invalid. Another approach called the study period is then preferred.

Example 4.3. Lowest Common Multiple (LCM) Approach. Continuing from the diaper production facility example, assume that the mutually exclusive production line alternatives have different expected lives and salvage values, as summarized in Table 4.1.

Alternative	Initial Cost (\$)	Annual Revenue (\$)	Salvage Value (\$)	Life Expected (years)
A (New line)	-6,000,000	1,825,000	500,000	6
B (Used line)	-2,500,000	730,000	200,000	3
C (Co-developed)	-5,000,000	1,095,000	400,000	4

Table 4.1: Summary of mutually exclusive alternatives for the diaper production facility.

Using a MARR of 10%, evaluate the alternatives using the LCM approach.

Solution: The expected lives are 6, 3, and 4 years. The Lowest Common Multiple (LCM) of these lives is 12 years. We evaluate the Present Worth of each alternative over a 12-year period. This is illustrated in Figure 4.4.

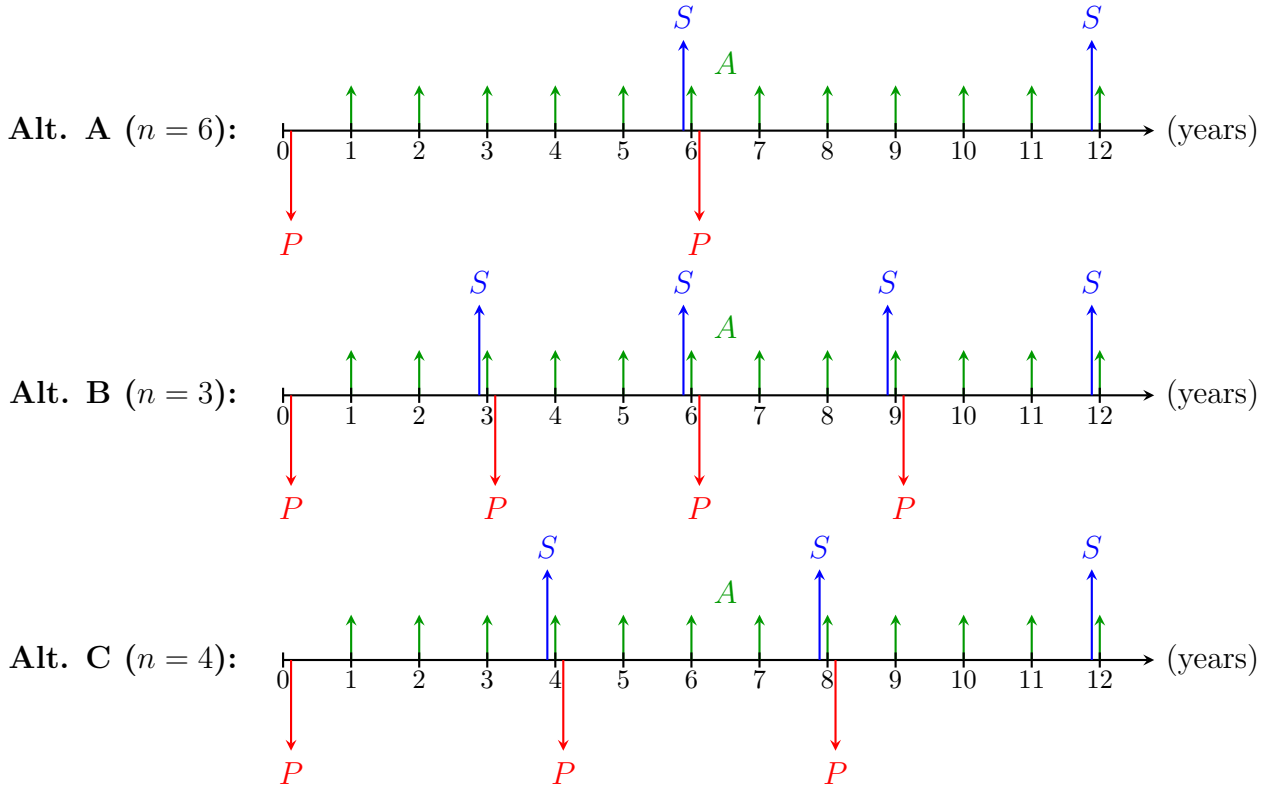


Figure 4.4: Cash flow timelines for mutually exclusive Alternatives A, B, and C over the 12-year LCM period.

For **Alternative A**, the initial cost is reinvested at $t = 6$, and salvage values are received at $t = 6$ and $t = 12$:

$$\begin{aligned}
 PW_A = & -6,000,000 - 6,000,000(P/F, 10\%, 6) \\
 & + 500,000(P/F, 10\%, 6) + 500,000(P/F, 10\%, 12) \\
 & + 1,825,000(P/A, 10\%, 12)
 \end{aligned}$$

$$= \$3,489,696.37$$

For **Alternative B**, reinvestment occurs at $t = 3$, $t = 6$, and $t = 9$, with salvage values at $t = 3$, $t = 6$, $t = 9$, and $t = 12$:

$$\begin{aligned} PW_B &= -2,500,000 - 2,500,000(P/F, 10\%, 3) \\ &\quad - 2,500,000(P/F, 10\%, 6) - 2,500,000(P/F, 10\%, 9) \\ &\quad + 200,000(P/F, 10\%, 3) + 200,000(P/F, 10\%, 6) \\ &\quad + 200,000(P/F, 10\%, 9) + 200,000(P/F, 10\%, 12) \\ &\quad + 730,000(P/A, 10\%, 12) \\ &= -\$1,464,017.41 \end{aligned}$$

For **Alternative C**, reinvestment occurs at $t = 4$ and $t = 8$, with salvage values at $t = 4$, $t = 8$, and $t = 12$:

$$\begin{aligned} PW_C &= -5,000,000 - 5,000,000(P/F, 10\%, 4) \\ &\quad - 5,000,000(P/F, 10\%, 8) + 400,000(P/F, 10\%, 4) \\ &\quad + 400,000(P/F, 10\%, 8) + 400,000(P/F, 10\%, 12) \\ &\quad + 1,095,000(P/A, 10\%, 12) \\ &= -\$2,699,350.97 \end{aligned}$$

Select Alternative A because it yields the highest positive Present Worth.

Important – Common Excel Trap: The NPV Function

When computing Present Worth (PW) in Excel, a common mistake is to pass the entire cash flow range (including Year 0) to the NPV function, e.g., `=NPV(rate, CF0:CFn)`.

Excel's NPV function strictly assumes that the first value in the range occurs at the *end of Period 1*. If you include the Year 0 cash flow inside the function, Excel incorrectly discounts the Year 0 cash flow by 1 period, Year 1 by 2 periods, etc. This effectively discounts the entire project by one extra year, meaning the computed result will be exactly your true *PW* divided by $(1 + i)$.

To correctly compute PW in Excel, you must evaluate the NPV starting from Period 1 and then manually add the non-discounted Year 0 cash flow:

$$PW = \text{NPV}(\text{rate}, \text{CF1:CFn}) + CF_0$$

Study Period and Alternative Life

A study period analysis is necessary if the first assumption about the length of time the alternatives are needed cannot be made. A time horizon is chosen over which the economic analysis is conducted, and only those cash flows which occur during that time period are considered relevant to the analysis. While individual cash flows occurring beyond the study period are excluded from the timeline, their remaining economic worth is not completely ignored. Instead, it is captured by assigning an estimated terminal (or market) value to the asset at the end of the study period.

Depending on the life of an alternative, three cases could occur when adopting the study period approach:

1. Alternative Life = Study Period: No adjustment to the cash flow is required.
2. Alternative Life > Study Period: A terminal value must be assigned to the alternative at the end of the study period. It is important to distinguish between three concepts often used to determine this value:
 - **Market Value:** What the asset can actually be sold for on the open market today (e.g., selling a 3-year-old truck to a dealer for \$15,000).
 - **Book Value:** The theoretical remaining value based on the remaining cash flows (e.g., the present worth of all expected cash flows beyond the study period is \$20,000).

- **Salvage Value:** The estimated value of the asset at the end of its intended useful life (e.g., selling the truck for scrap metal for \$2,000 after 10 years).

For the study period approach, this terminal value is typically based on the *market value* of the asset at that specific point in time, or its *book value*.

3. Alternative Life < Study Period: Assumptions must be made on what happens in the additional years based on the cash flow type:

- Service (Cost) Alternative: Estimate the cost of continuing service over the additional years.
- Revenue Alternative: Reinvest all cash flows in other opportunities available to the firm at the MARR to the end of the study period.

Important – Study Period Principle for Mutually Exclusive Alternatives

The underlying principle for comparing mutually exclusive alternatives is to have the same study period.

Note – Study Period Quick Reference

The table below summarizes the required adjustment for each case.

Case	Required Adjustment
Alt. Life = Study Period	No adjustment required.
Alt. Life > Study Period	Estimate a terminal market (or book) value at the end of the study period and treat it as a cash inflow.
Alt. Life < Study Period	Compound all net cash flows at the end of the alternative's life forward to the end of the study period at the MARR.

Example 4.4. Study Period Approach.

Reconsider the diaper production facility alternatives from Table 4.1. Suppose the company wishes to compare the alternatives over a fixed **5-year study period**. A minimum attractive rate of return (MARR) of 10% is used.

Since Alternative A has a useful life longer than the study period (6 years > 5 years), its market value at the end of Year 5 must be estimated. Assume its estimated market value at Year 5 is \$1,000,000.

Alternatives B and C have useful lives shorter than the study period (3 and 4 years, respectively). Under the **study period approach**, these alternatives are *not replaced* when they reach the end of their useful lives. Instead, all accumulated net cash flows at the end of each alternative's life are assumed to be invested at the MARR until the end of the study period.

Evaluate the alternatives using the Present Worth method.

Solution:

For **Alternative A**, the analysis is truncated at the end of Year 5. The estimated market value at Year 5 acts as the terminal (salvage) value.

$$\begin{aligned} PW_A &= -6,000,000 + 1,000,000(P/F, 10\%, 5) + 1,825,000(P/A, 10\%, 5) \\ &= \$1,539,107.18 \end{aligned}$$

Important – Study Period Rule for Shorter-Lived Alternatives

When an alternative has a useful life shorter than the study period, it is **not replaced**. Instead:

1. Compute the future worth of all net cash flows over the alternative's life at the end of its useful life.
2. Compound this amount at the MARR until the end of the study period.
3. Discount the resulting amount back to the present and subtract the initial investment.

When using the $(F/A, i, n)$ factor, the future worth is computed *at the end of Year n* . Therefore, the final annual cash flow occurs at Year n and does not earn additional interest before that point, which is consistent with the timing of end-of-period cash flows.

For **Alternative B**, the machine reaches the end of its useful life after Year 3. The accumulated net cash flows at Year 3 are first computed, then invested at the MARR until Year 5.

$$\begin{aligned} FW_{\text{net},3} &= 730,000(F/A, 10\%, 3) + 200,000 \\ &= \$2,616,300.00 \end{aligned}$$

$$\begin{aligned}
 FW_{\text{net},5} &= FW_{\text{net},3}(F/P, 10\%, 2) \\
 &= 2,616,300(1.21) \\
 &= \$3,165,723.00
 \end{aligned}$$

$$\begin{aligned}
 PW_B &= -2,500,000 + FW_{\text{net},5}(P/F, 10\%, 5) \\
 &= -2,500,000 + 3,165,723(P/F, 10\%, 5) \\
 &= -\$534,335.09
 \end{aligned}$$

For **Alternative C**, the machine reaches the end of its useful life after Year 4. The accumulated net cash flows at Year 4 are invested for one additional year before being discounted to the present.

$$\begin{aligned}
 FW_{\text{net},4} &= 1,095,000(F/A, 10\%, 4) + 400,000 \\
 &= \$5,481,895.00
 \end{aligned}$$

$$\begin{aligned}
 FW_{\text{net},5} &= FW_{\text{net},4}(F/P, 10\%, 1) \\
 &= 5,481,895(1.10) \\
 &= \$6,030,084.50
 \end{aligned}$$

$$\begin{aligned}
 PW_C &= -5,000,000 + FW_{\text{net},5}(P/F, 10\%, 5) \\
 &= -5,000,000 + 6,030,084.50(P/F, 10\%, 5) \\
 &= -\$1,255,791.95
 \end{aligned}$$

Since Alternative A has the highest present worth, it remains the preferred alternative over the 5-year study period.

Important – A Good Habit

Whenever solving an engineering economy problem, begin by sketching the cash-flow diagram. The diagram clarifies the timing of each cash flow, makes equivalent cash-flow transformations easier to understand, and provides a convenient way to verify that the selected interest factors are applied correctly.

Note – Spreadsheet Precaution

While spreadsheets simplify calculations, their financial functions often make hidden assumptions about cash-flow timing. Always sketch the cash-flow diagram first to prevent misapplying them.

4.3 Future Worth (FW) Analysis

As demonstrated in the study period approach from the previous section, calculating the accumulated wealth of cash flows at a future date can be a practical way to evaluate projects. Future Worth (FW) Analysis is conceptually similar to Present Worth analysis, but it focuses on equivalent future values instead of present values. Like PW analysis, the MARR is used to compound these cash flows to the future.

The FW and PW criteria are mathematically equivalent in comparing alternatives; if an alternative is preferred using PW, it will also be preferred using FW. However, FW analysis is specifically utilized when:

- A primary goal of the firm is to maximize the future wealth of its stockholders.
- An asset may be sold after a specific startup period (e.g., acquiring a company to build its value and sell it in three years).
- Projects will not “come online” or start generating returns until the end of the investment period, such as in large-scale construction projects.

The FW of all cash flows is found by compounding each to period N :

$$FW(i\%) = \sum_{k=0}^N F_k (1+i)^{N-k}$$

Note that $FW = PW \times (F/P, i\%, N)$, which confirms the mathematical equivalence of both criteria.

Important – Future Worth Selection Guidelines

The selection guidelines for FW analysis are the same as for PW analysis; $FW \geq 0$ means the MARR is met or exceeded. For two or more mutually exclusive alternatives, select the one with the numerically largest FW value.

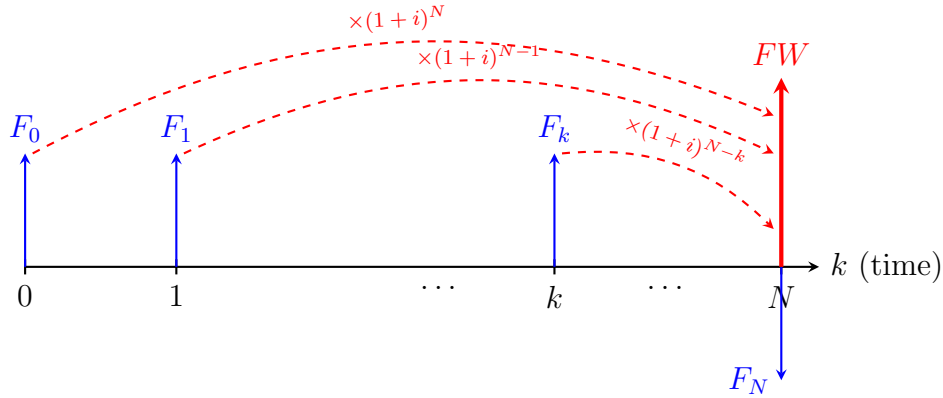


Figure 4.5: Illustrating the process of compounding future cash flows forward to time N to find the Future Worth.

4.4 Capitalized Cost Analysis

Many public sector projects such as bridges, dams, highways and toll roads, railroads, and hydroelectric and other power generation facilities have **very long expected useful lives**. Permanent endowments for charitable organizations and universities also have infinite lives.

The economic worth of these types of projects or endowments is evaluated using the present worth of the cash flows. For these projects, the life cycle, n , is either very long, indefinite, or infinity.

Definition – Capitalized Cost (CC)

Capitalized Cost (CC) is the **present worth** of a project that has a very long life (more than, say, 35 or 40 years) or when the planning horizon is considered very long or infinite.

Recall that the present worth given a uniform annuity A is:

$$\begin{aligned}
 P &= A(P/A, i\%, n) \\
 &= A \left(\frac{(1+i)^n - 1}{i(1+i)^n} \right) \\
 &= A \left(\frac{1 - (1+i)^{-n}}{i} \right)
 \end{aligned} \tag{4.1}$$

The CC for an infinite uniform series of cash flows (with annuity A) is found by

taking the limit of 4.1 as $n \rightarrow \infty$:

$$\begin{aligned} CC &= \lim_{n \rightarrow \infty} A \left(\frac{1 - (1 + i)^{-n}}{i} \right) \\ &= \frac{A}{i} \end{aligned} \quad (4.2)$$

4.4.1 Evaluation of CC for Any Cash Flow

The procedure to determine the CC for any cash flow is:

1. Draw a cash flow diagram showing all non-recurring (one-time) cash flows and at least two cycles of all recurring (periodic) cash flows³.
2. For non-recurring (one-time) cash flows: the CC is their corresponding PW.
3. For a recurring cash flow of value R that repeats every n_R years: Find the equivalent uniform annual worth through one life cycle of recurring amounts:

$$A_R = R(A/F, i, n_R)$$

4. Find equivalent CC for the A_R series:

$$CC_R = \frac{A_R}{i}$$

Example 4.5. American University of Beirut (AUB) Software System. The American University of Beirut (AUB) has decided to install a new campus-wide software system called **Woohoo**. The system has an expected infinite life. Determine the Capitalized Cost (CC) of the software given an interest rate of 5% per year and the following cost estimates:

- Initial installed cost: \$150,000
- One-time hardware upgrade cost after 10 years: \$50,000
- Annual maintenance contract: \$5,000 for the first 4 years, and \$8,000 thereafter (years 5 to ∞)
- Recurring major software upgrade cost: \$15,000 every 13 years

³Drawing the cash flow diagram is more important in CC calculations than elsewhere, because it helps separate non-recurring and recurring amounts.

We first draw the combined cash flow diagram for this project over two lifecycles ($N = 26$ years):

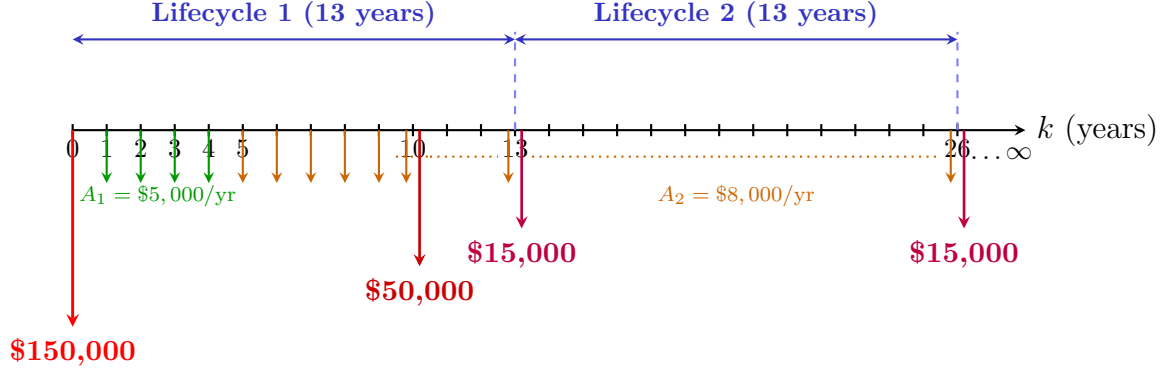


Figure 4.6: Combined cash flow diagram for the Woohoo software system showing non-recurring costs, maintenance, and two full lifecycles of major software upgrades ($N = 26$).

We apply the procedure to find the CC for the given cash flows at $i = 5\%$.

Step 1 & 2: Non-recurring cash flows The initial cost occurs at $t = 0$ and the hardware upgrade at $t = 10$. We find their present worth:

$$\begin{aligned} CC_{\text{non-recurring}} &= -150,000 - 50,000(P/F, 5\%, 10) \\ &= -150,000 - 50,000(0.61391) \\ &= -\$180,695.66 \end{aligned}$$

Step 3 & 4: Recurring cash flows We divide the recurring costs into the maintenance contract and the software upgrades.

Maintenance Contract: The cost is \$5,000 annually for years 1 through 4, and \$8,000 annually from year 5 to infinity. We can break this down into two concurrent infinite series:

1. A base infinite annual cost of \$5,000 starting at year 1.
2. An additional perpetual annual cost of \$3,000 that only starts at year 5.

First, we find the capitalized cost of the base \$5,000 series at $t = 0$:

$$CC_{\text{base}} = \frac{-5,000}{0.05} = -100,000$$

Second, we find the capitalized cost of the additional \$3,000 series. Since it starts in

year 5, dividing by i gives us its value one period prior, at $t = 4$:

$$CC_{\text{extra at } t=4} = \frac{-3,000}{0.05} = -60,000$$

We must then discount this value back to $t = 0$ using the P/F factor:

$$CC_{\text{extra at } t=0} = -60,000(P/F, 5\%, 4) = -60,000(0.82270) = -49,362.15$$

Adding them together yields the total maintenance CC:

$$\begin{aligned} CC_{\text{maintenance}} &= CC_{\text{base}} + CC_{\text{extra at } t=0} \\ &= -100,000 - 49,362.15 \\ &= -\$149,362.15 \end{aligned}$$

Major Software Upgrade: The upgrade costs \$15,000 every $n_R = 13$ years. We first convert this to an equivalent uniform annual worth (A_R):

$$\begin{aligned} A_R &= -15,000(A/F, 5\%, 13) \\ &= -15,000(0.05646) \\ &= -\$846.84 \end{aligned}$$

Then, we find the capitalized cost for this infinite series:

$$\begin{aligned} CC_{\text{upgrade}} &= \frac{A_R}{0.05} = \frac{-846.84}{0.05} \\ &= -\$16,936.73 \end{aligned}$$

This transformation process is illustrated in Figure 4.7.

Total Capitalized Cost:

$$\begin{aligned} CC_{\text{total}} &= CC_{\text{non-recurring}} + CC_{\text{maintenance}} + CC_{\text{upgrade}} \\ &= -180,695.66 - 149,362.15 - 16,936.73 \\ &= -\$346,994.54 \end{aligned}$$

The total capitalized cost of the Woohoo software system is \$346,994.54.

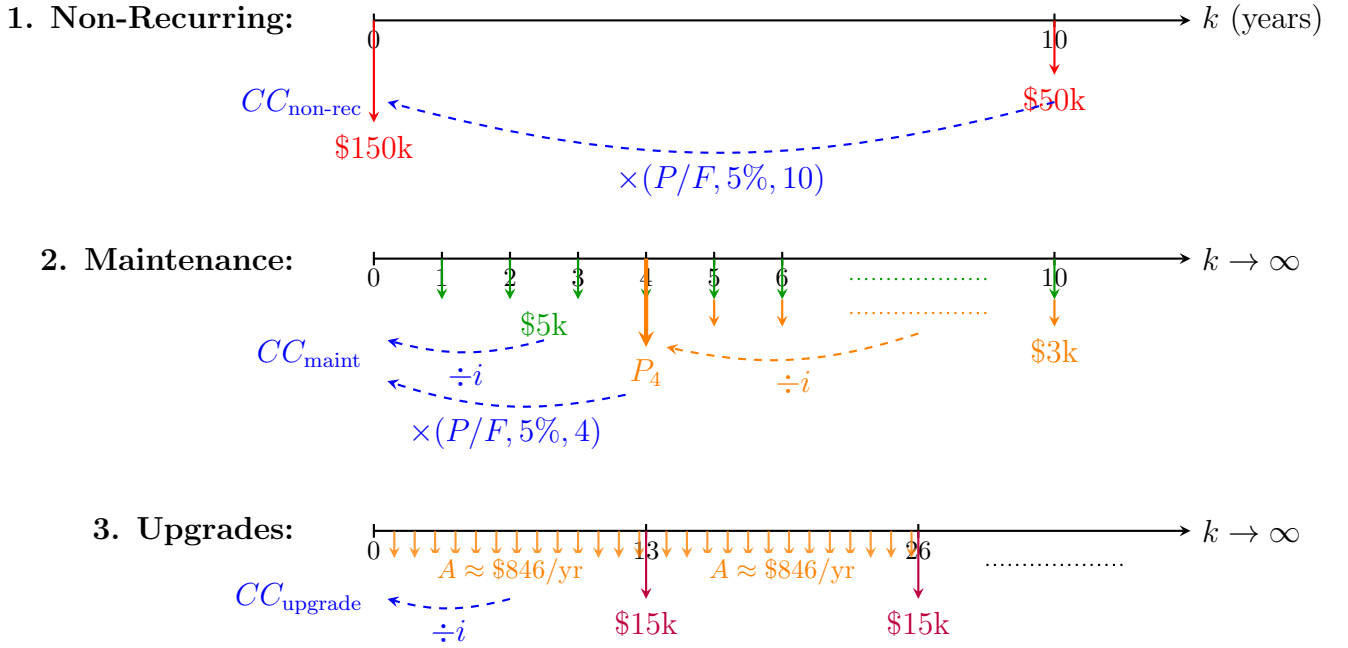


Figure 4.7: Visual transformation of the non-recurring, maintenance, and upgrade cash flows into Capitalized Cost at $t = 0$.

4.4.2 Comparing Alternatives

Alternatives that have infinite lives can be directly compared on the basis of their Capitalized Cost (CC), which is mathematically equivalent to the PW criteria over an infinite time horizon.

If a finite-life alternative (for example, an asset with a 5-year life) is compared to one with an indefinite or very long life, capitalized costs can be used to evaluate both over an **infinite study period**⁴. To determine the capitalized cost for the finite-life alternative, simply calculate the equivalent uniform annual worth (A) for one life cycle and divide by the interest rate:

$$CC_{\text{finite}} = \frac{A}{i}$$

4.5 Payback Period Analysis

Payback analysis is another use of the present worth technique.

⁴In fact, this is an application of the Study Period approach where the assigned study period is infinity. The finite alternative is thus extended to an infinite time horizon.

Definition – Payback Period

The payback period n_P is an estimated time for the revenues, savings, and any other monetary benefits to completely recover the initial investment **plus** a stated rate of return i .

Given that the Net Cash Flow (NCF) is:

$$NCF_t = \text{cash inflows at time } t - \text{cash outflows at time } t$$

the payback period n_P is found by solving:

$$0 = -P + \sum_{t=1}^{n_P} NCF_t(P/F, i, t)$$

where P is the initial investment. This equation can be solved using trial and error or using a computer package, such as Excel.

If $i = 0\%$, then $0 = -P + \sum_{t=1}^{n_P} NCF_t$. Additionally, if the NCF is uniform, that is $NCF_t = NCF$, then

$$n_P = \frac{P}{NCF}$$

Important – Caution on Payback Period Analysis

Payback period analysis is primarily used in practice as a **quick initial screening method** or a supplemental tool, rather than as the primary means for making accept/reject decisions. Caution is required because:

- **Disregards subsequent cash flows:** It completely ignores all cash flows that occur *after* the payback period n_P , even if those cash flows significantly increase the total return on the investment.
- **Neglects time value of money (if $i = 0\%$):** No-return payback calculations fail to account for the time value of money, as no return on the investment is required.

For example, if a machine has a planned use of only 3 years but its calculated payback is 6 years, it strongly indicates the equipment should not be obtained. However, even this 6-year payback figure should only be considered supplemental information and cannot replace a complete economic analysis.

Example 4.6. The Trap of Payback Period Analysis. Consider two mutually exclusive machines being evaluated for a 5-year project using a 0% payback method.

- **Machine X:** Costs \$10,000, generates \$2,500/year for 5 years. Salvage value is \$0.
- **Machine Y:** Costs \$10,000, generates \$2,000/year for 5 years. Salvage value is \$8,000 at year 5.

The 0% payback period (n_P) is:

- $n_{P,X} = \frac{10,000}{2,500} = 4$ years
- $n_{P,Y} = \frac{10,000}{2,000} = 5$ years

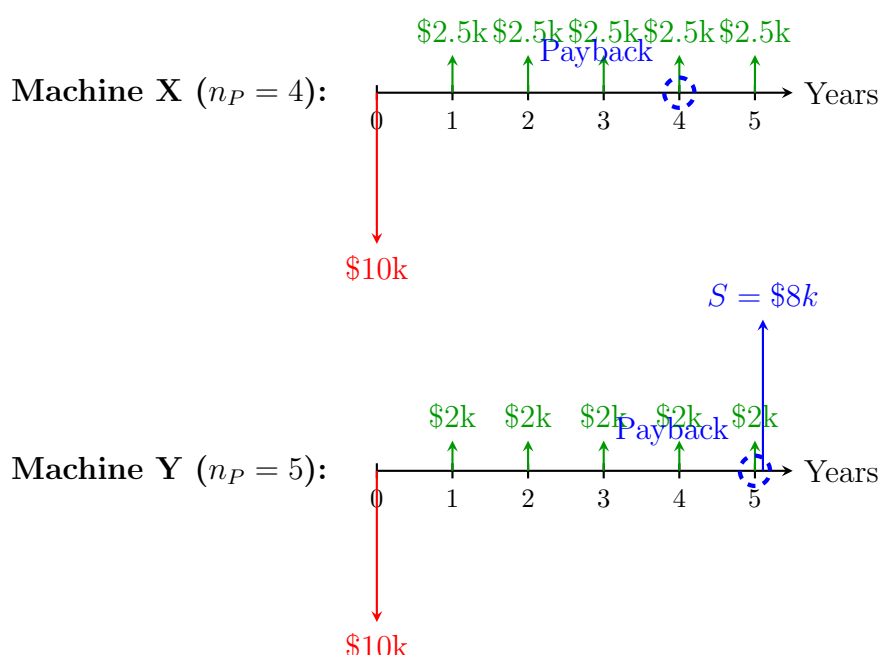


Figure 4.8: Cash flow diagram illustrating how Machine X reaches payback earlier, but Machine Y yields significantly more total return due to the ignored salvage value.

Based on payback analysis, Machine X is preferred because it recovers the investment faster. However, examining the cash flow diagram in Figure 4.8 makes it obvious that Machine Y generates more total wealth (\$18,000 total vs \$12,500 total) simply by waiting one more year for the salvage value. Payback period analysis completely blinds the decision maker to the impact of future cashflows.

4.6 Life-Cycle Costs

Definition – Life-Cycle Cost (LCC) Analysis

Life-cycle cost (LCC) analysis utilizes Annual Worth (AW) or PW methods to evaluate cost estimates for the entire life cycle of one or more projects. Estimates will cover the entire life span from the early conceptual stages, through the design and development stages, through the operating stages, and even the phaseout and disposal stages. Both direct and indirect costs are included to the extent possible, and differences in revenue and savings projections between alternatives are included.

4.6.1 Phases of an Engineering Project

Generally, the LCC estimates may be categorized into a simplified format for the major phases of *acquisition*, *operation*, and *phaseout/disposal*, and their respective stages.

- **Acquisition Phase:** Includes all activities prior to delivery (commonly 75–85% of total cost).
 - *Requirements Definition:* Determining user needs and system requirements.
 - *Preliminary Design:* Feasibility studies, early planning, and making final go/no-go decisions.
 - *Detailed Design:* Detailed resource planning and acquiring assets.
- **Operation Phase:** Activities while the system is functioning.
 - *Construction and Implementation:* Purchasing, building, and testing components.
 - *Usage:* Generating products and services; typically the longest portion of the life cycle.
- **Phaseout and Disposal Phase:** Transitioning to a new system, including removal and recycling.

Note – Importance of Disposal Phase from a Sustainability Perspective

This phase is becoming increasingly important as it directly impacts the sustainability of the project. Additionally, planning for a sustainable phase-out has a tremendous impact on decisions made during the initial Acquisition Phase.

Cost Distribution and Commitment Over the Life Cycle

Understanding how costs are distributed and committed over time is a critical concept in engineering economy.

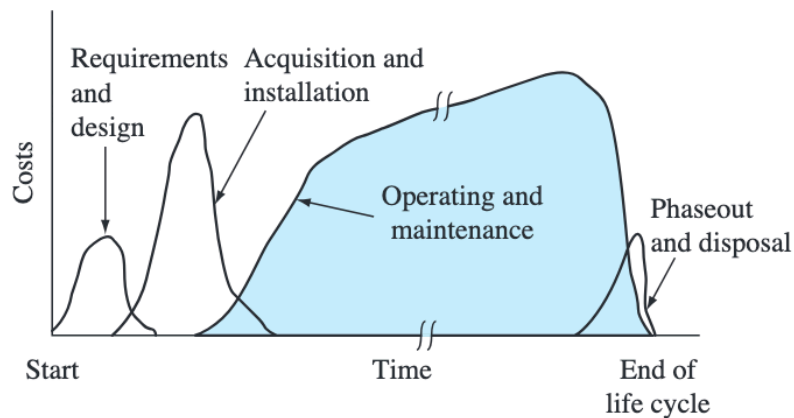


Figure 4.9: Typical distribution of costs over the life cycle phases. source: blank'engineering'2012

As illustrated in Figure 4.9, costs do not occur evenly. For some systems (e.g., defence and aviation), operating and maintenance costs ramp up quickly after the acquisition phase and remain high until phaseout.

Committed vs. Actual Costs

Definition – Committed vs. Actual Costs

Committed Cost: The cumulative cost locked in by early design, engineering, and architectural decisions (even though funds have not yet been disbursed).

Actual Cost: The actual expenditure or disbursement of funds.

Key Takeaway: Typically, **75% to 85%** of the total Life Cycle Cost (LCC) is **committed** during the preliminary and detailed design stages, long before major actual spending takes place.

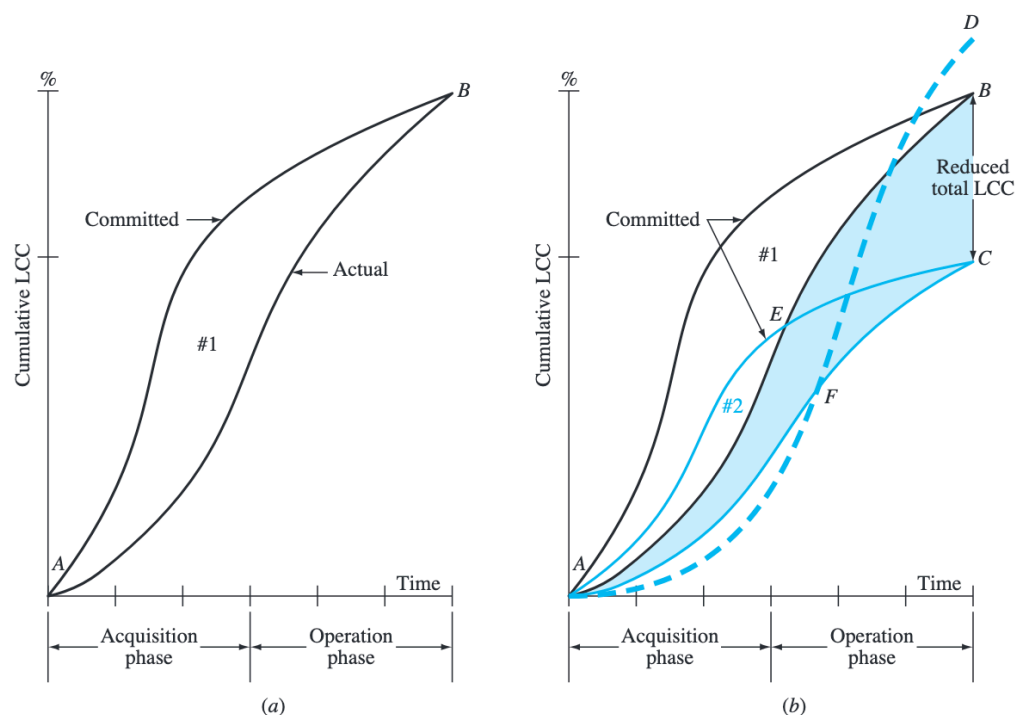


Figure 4.10: Committed vs. Actual LCC. (a) Base scenario where actual cost trails committed cost. (b) Alternative scenarios showing reduced LCC through early design, and increased LCC due to ad-hoc savings. source: **blank'engineering'2012**

As shown in Figure 4.10(a), the *actual* cost curve trails the *committed* cost curve. Because so much cost is committed upfront, the **greatest opportunity to significantly reduce the total LCC lies in the early design stages**. By investing in better design and efficient equipment early on, you can lower the entire committed cost curve, which inevitably reduces the actual costs incurred later. This creates a more effective curve (Design #2), as highlighted by the shaded region in Figure 4.10(b).

Important – The Trap of Short-Term Cost Savings

Management sometimes introduces unplanned cost-cutting measures during the acquisition or construction phases (for example, using inferior-strength materials to save money upfront).

While this may appear to generate savings, these decisions frequently backfire. As depicted by the dashed line extending upwards in Figure 4.10(b), they **substantially increase costs later** during the usage stage due to structural failures, excessive maintenance, or system inefficiencies. When evaluating projects, always prioritize the total life cycle cost over short-term, initial savings.